

DNS

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Summary:

- **Normal DNS:** Low volume, simple domains, UDP port 53, expected record types.
- **Malicious DNS:** High-entropy subdomains, periodic beaconing, tunneling via long queries, or DNS over TCP for small data.
- **Detection workflow:**
 1. Capture traffic (**Wireshark** / **tcpdump** / **Zeek**).
 2. Filter DNS (dns in Wireshark).
 3. Check entropy, volume, and patterns.
 4. Correlate with host logs (processes making DNS requests).
 5. Investigate root cause and block suspicious domains.

Common Tools & Commands:

- **Wireshark:** Filter dns, inspect queries/responses.
- **tcpdump:** tcpdump -nn port 53.
- **Zeek:** Process PCAP for dns.log and entropy analysis.
- **Hosts file check:** Look for unexpected entries.

1. DNS Basics

What DNS does: Translates human-readable domain names (e.g., www.example.com) into IP addresses (e.g., **93.184.216.34**).

Query/Response Flow:

- Client sends **DNS query** (usually over **UDP** port 53).
- Resolver (local or recursive server) looks up the record.
- Server returns **DNS response** with the IP (A record) or other data.

Common Record Types:

- **A** — IPv4 address.
- **AAAA** — IPv6 address.
- **CNAME** — alias to another name.
- **MX** — mail server.
- **TXT** — text (often used for verification or malicious data).

Security relevance: DNS is trusted and rarely blocked. Attackers abuse it for:

- **C2 communication** (malware queries attacker-controlled domains).
- **DNS tunneling** (hide data inside DNS queries/responses).
- **Exfiltration** (encode stolen data in subdomains).

Normal vs. Malicious:

- Normal: Short, simple queries; low volume; standard record types.
- Malicious: High volume, high-entropy subdomains, unusual record types, or DNS over TCP (rare for normal use).

Key takeaway: DNS is a common covert channel. Monitor for anomalies in query volume, subdomain patterns, and entropy.

2. Creating and Inspecting Normal DNS Traffic

Generating traffic (demo approach):

- Use browser or tools like **curl** to trigger DNS queries.
- Capture with **Wireshark** or **tcpdump** on the interface.

Example tcpdump command:

```
Bash
tcpdump -nn -i eth0 port 53
```

Inspecting in Wireshark:

- Filter: dns
- Look at:
 - **Query** section: Domain name, record type (A, AAAA, etc.).
 - **Response** section: Resolved IP, TTL.

UDP vs TCP for DNS:

- **UDP** (default, port 53): Fast, connectionless — used for most queries.
- **TCP** (port 53): Used for large responses (e.g., zone transfers) or when UDP is blocked.
- Malicious indicator: DNS over TCP for small queries → possible tunneling.

Key takeaway: Baseline normal DNS (low volume, simple domains). Any deviation (high volume, encoded subdomains) is suspicious.

3. Modifying Traffic Flow & Hosts File

Hosts File:

- Local file that overrides DNS (maps domain to IP manually).
- Location: C:\Windows\System32\drivers\etc\hosts (Windows) or /etc/hosts (Linux).

Security relevance:

- Attackers modify hosts file for redirection (phishing, C2 hiding).
- Detection: Unexpected entries pointing to suspicious IPs.

Demo insight:

- Modifying hosts file changes resolution without querying real DNS.
- Useful for testing or attacker persistence (bypass DNS monitoring).

Key takeaway: Check hosts file during host forensics — it can bypass network DNS controls.

4. Detecting Malicious DNS (C2 and Tunneling)

DNS C2:

- Malware uses DNS queries to communicate with attacker (beaconing, commands in subdomains).
- Indicators: Periodic queries to random-looking domains, high entropy in subdomains.

DNS Tunneling:

- Hide data inside DNS queries/responses (encode payloads in subdomains).
- Indicators: Very long subdomains, high entropy, unusual volume, or TXT records with large data.

Detection Techniques:

- **Entropy score:** High randomness in subdomain names (use **Zeek** scripts).
- Volume: Sudden spike in DNS queries from one host.
- Pattern: Queries without corresponding normal web traffic.

Demo insight (Generating Entropy Score with Zeek):

- **Zeek** processes PCAP and generates logs (dns.log).
- Look for high-entropy subdomains or unusual query types.
- Command example (simplified):

```
Bash
zeek -r capture.pcap
```

- Review dns.log for suspicious entries.

Advanced Analysis:

- Correlate DNS with host processes (e.g., malware making repeated queries).
- Baseline normal DNS behavior in your environment.
- Use Suricata rules or SIEM to alert on high-entropy or anomalous DNS.

Key takeaway: DNS is stealthy. Look for **high entropy**, unusual volume, or DNS without normal application traffic as strong C2/tunneling indicators.